



Class: Final Year B.Tech.

Course: Advanced
Construction Materials

Course Code: PEC-CE415

Date: 14.10.2025

Time: 14:30 to 15:15

Max. Marks: 20

Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- Figures to right indicate full marks.

Q.N.			Marks	CO	BT
Q.1	a	What are the different coating methods of reinforcement? Explain Fusion Bonded Polyurethane (FBPU) coating with a neat labelled diagram.	05	CO2	BT3
	b	What are the mechanical characteristics of Polymer? Plot a neat labelled graph showing tensile response of all three failure.	05	CO3	BT3
Q.2	a	Define polymer and list out the various types of polymers. Describe in detail polymers classified according to their thermal behaviour.	04	CO2	BT2
	b	What is ceramic composite and give its four applications.	03	CO3	BT3
Q.3	a	Differentiate between veneers and laminates with respect to their cost, Environmental Impact, and Repairability.	03	CO2	BT2

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyse, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Understand basics of building construction and concrete technology
CO 2	Understand properties and its utilities of concrete, metals and various composites, admixtures, smart and intelligent materials
CO 3	Apply concrete, metals and various composites, admixtures, smart and intelligent materials. For various construction systems.
CO 4	Analyse the performance various forms of concrete, metals, composites, admixtures, smart and intelligent materials



IN-SEMESTER EXAMINATION - II (ISE-II) - 2025-26 (Semester-I)

Date: Tuesday, October 14, 2025

Course: Bridge Analysis and Design

Time: 14:30 - 15:15

Class: Final Year Civil Engineering

Course Code: PEC-CE412

Maximum Marks: 20

- Note**
1. Attempt all questions.
 2. Use of non-programmable calculator is permitted.
 3. Draw appropriate figures wherever required.
 4. Assume appropriate data, if necessary.
 5. Clearly specify / write assumptions considered.

Que. No.	Question statement	Mark	CO	BT
	A bridge is to be built on a river. The length of bridge is 64 m with each span of 16 m. The bridge is to have two lanes of 3.75 m. The deck is to be provided with kerb on both sides of 600 mm width and shall be raised by 300 mm from road top level. It was decided to provide a Tee-girder bridge for this configuration. The slab thickness is 200 mm and girder depth as 1400 mm. Further, three girders are to be provided at centre-to-centre distance of 2.5 m. Consider 5 cross girder to be provided of same depth as that of main girder.			
01	Draw the position of the IRC class AA tracked vehicle on bridge deck with neat dimension of placed vehicle (not to scale diagram acceptable) for maximum bending moment on bridge members as per IRC specifications.	04	CO2	BT3
02	Determine the moment on the slab for the design of slab (Consider live load only).	06	CO4	BT4
03	Determine the moment on to longitude girder for maximum moment (Consider only live load including impact factor). Suggest the design moment on the each main (longitude) girder employing Courbons method for load analysed with suitable factor.	10	CO3	BT4

Course Outcome (CO)	Bloom's Taxonomy Level (BT)
CO1 To recall the fundamentals of water resources engineering.	BT1 - Remember
CO2 To understand the components of bridges, its classifications and importance.	BT2 - Understand
CO3 To apply various theories of lateral load distribution for component design of bridge.	BT3 - Apply
CO4 To analyse any given type of bridge.	BT4 - Analyse
CO5 To get exposure to evaluation of sub structures, type of foundations, importance of bearings, lessons from bridge failures.	BT5 - Evaluate
	BT6 - Create

* * * All the Very Best * * *

SUPPLYMENTARTY DATA REQUIRED AS PER THE RELEVANT IRC codes and formulae:

$$J \approx \frac{ab^3}{16} \left(\frac{16}{3} - 3.36 \frac{b}{a} \left(1 - \frac{b^4}{12a^4} \right) \right)$$

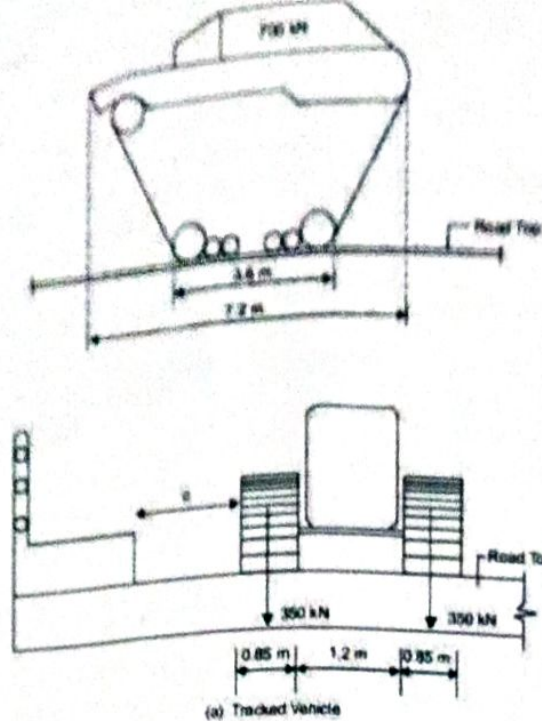
where

a is the length of the long side

b is the length of the short side

Values of Torsion Coefficient R

b/a	R	b/a	R	b/a	R
1	0.141	2.25	0.24	5	0.291
1.2	0.166	2.5	0.249	10	0.312
1.5	0.196	3	0.263		
2	0.229	4	0.281		



(1) **Solid slab spanning in one direction**

- (i) For a single concentrated load, the effective width may be calculated in accordance with the following equation:

$$b_{ef} = \alpha \cdot a \left(1 - \frac{a}{l_o} \right) + b_1$$

Eq. B3.1

b/l_o	α for simply supported slab	α for continuous slab	b/l_o	α for simply supported slab	α for continuous slab
0.1	0.40	0.40	1.1	2.60	2.28
0.2	0.80	0.80	1.2	2.64	2.36
0.3	1.16	1.16	1.3	2.72	2.40
0.4	1.48	1.44	1.4	2.80	2.48
0.5	1.72	1.68	1.5	2.84	2.48
0.6	1.96	1.84	1.6	2.88	2.52
0.7	2.12	1.96	1.7	2.92	2.56
0.8	2.24	2.08	1.8	2.96	2.60
0.9	2.36	2.16	1.9	3.00	2.60
1.0	2.48	2.24	2 & above	3.00	2.60

7.6.1.2 **Effective width of flanged beams and box beams**

- (1) For analysis of section for ULS & SLS effective width shall be taken as given in Fig. 7.1 & Eq. 7.1.

The effective flange width b_{ef} for a T beam or L beam may be derived as:

$$b_{ef} = \sum b_{ef,i} + b_w \leq b$$

Eq. 7.1

with $b_{ef,i} = 0.2b_i + 0.1l_o \leq 0.21l_o$,

and $b_{ef,i} \leq b_i$

(For the notations see Fig. 7.1)

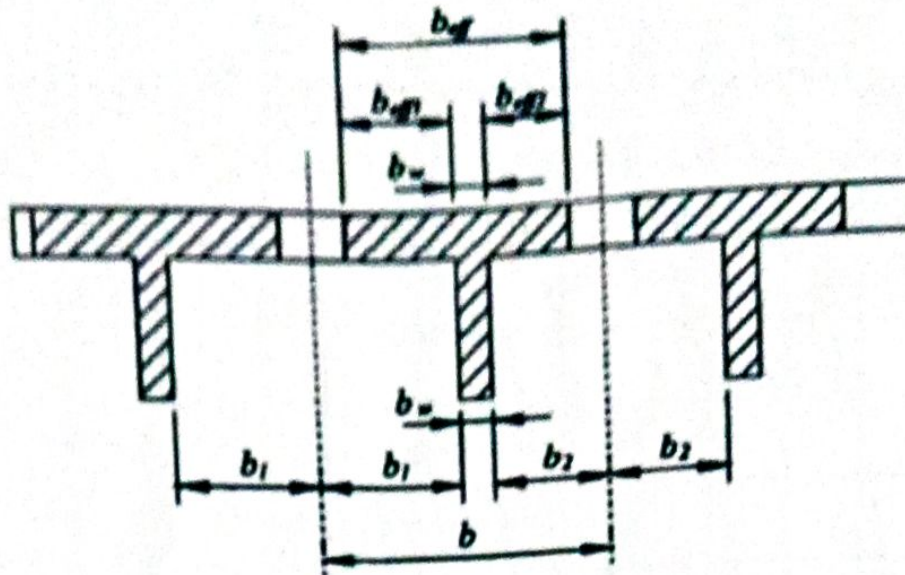


Fig. 7.1 Definition of Parameters to Determine Effective Flange Width

- (2) l_e = The distance between the points of zero moments (in the absence of rigorous calculations for continuous span it may be taken as 0.7 times effective span). The effective width may be taken as constant for the full span.
- (3) For limit state check of vibration for footbridge the actual flange width may be used.

10.3.2 Elements not requiring design shear reinforcement

- (1) The design shear resistance $V_{Rd,c} \geq V_{Ed}$
- (2) The design shear resistance of the member without shear reinforcement $V_{Rd,c}$ is given by:

$$V_{Rd,c} = \left[0.12K(80\rho_1 \cdot f_{ck})^{0.33} + 0.15\sigma_{cp} \right] b_w d$$

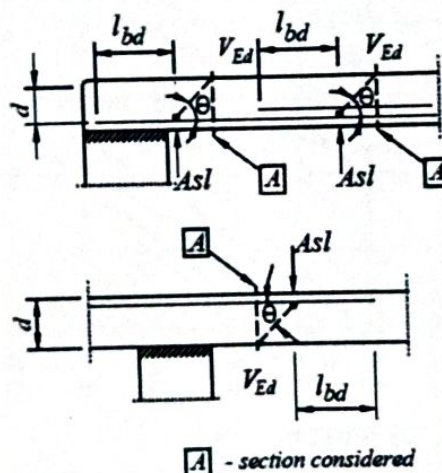
Subject to a minimum of $V_{Rd,c} = (v_{min} + 0.15\sigma_{cp}) b_w d$ Eq. 10.1

$$K = 1 + \sqrt{\frac{200}{d}} \leq 2.0 \text{ where } d \text{ is depth in millimeters.} \quad \text{Eq. 10.2}$$

$$\text{and } v_{min} = 0.031K^{3/2} f_{ck}^{1/2} \quad \text{Eq. 10.3}$$

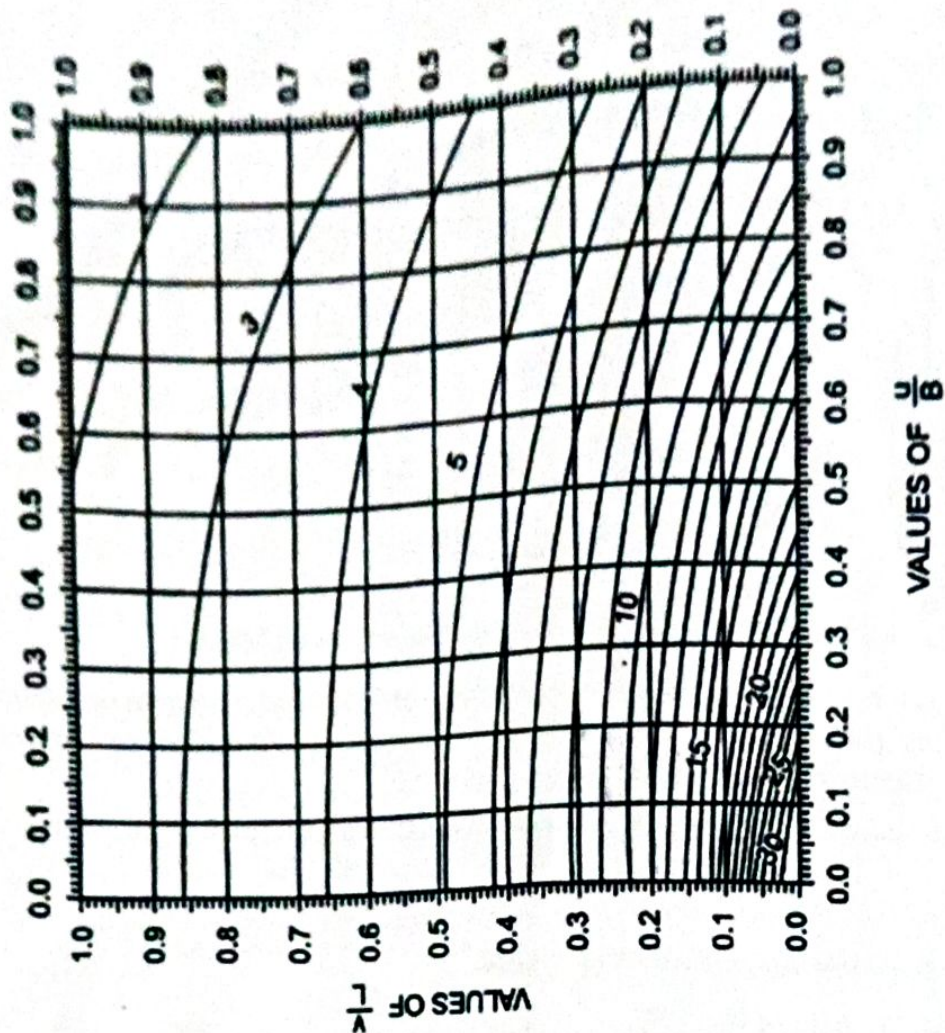
σ_{cp} is limited to $0.2 f_{cd}$ (MPa) where $\sigma_{cp} = N_{Ed} / A_c < 0.2 f_{cd}$ (MPa)

$$\rho_1 = \frac{A_{sl}}{b_w d} \leq 0.02 \quad (\text{Fig. 10.5})$$

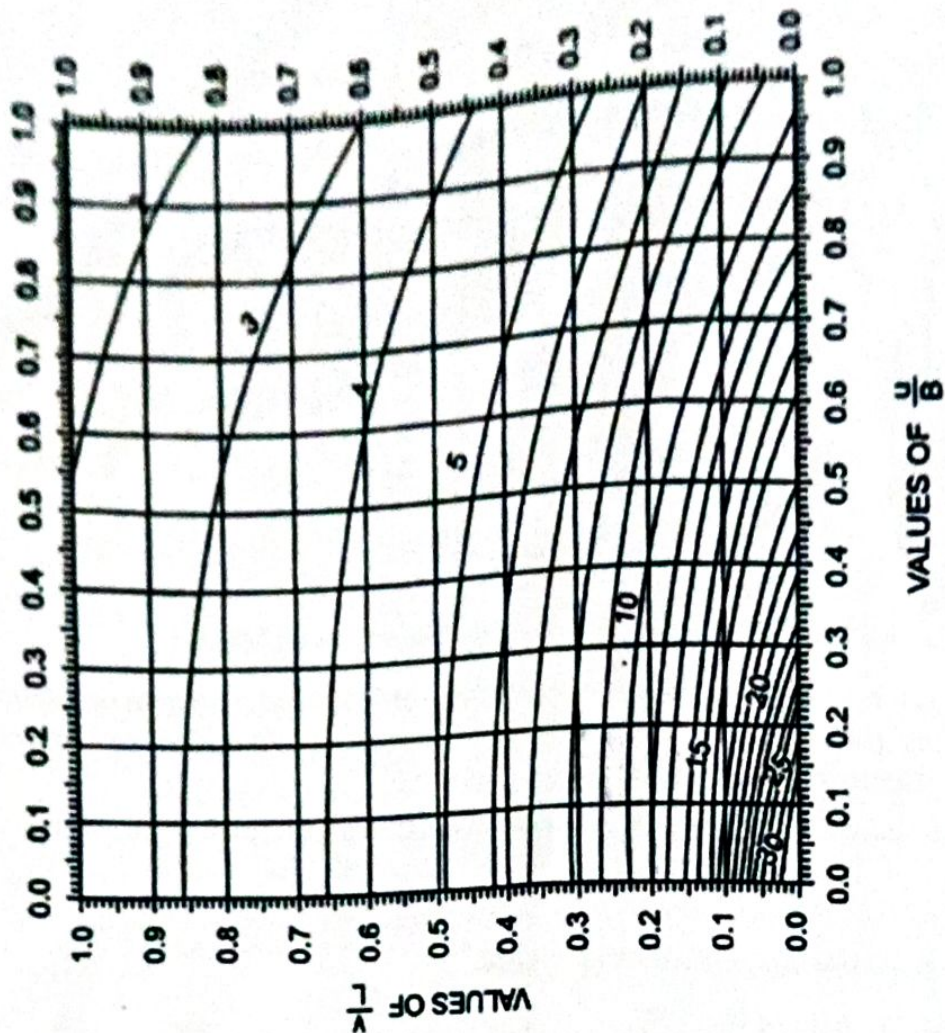


[A] - section considered

Fig. 10.5 Definition of A_{sl} in Expression ρ_1



a. COEFFICIENT $m_1 \times 100$



b. COEFFICIENT $m_2 \times 100$

Pigeaud's Curves of coefficient m_1 and m_2 for $K = 0.6$



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Civil Engineering
In-Semester Examination II
Academic Year: 2025-26 (Semester – I)

Class: Final Year
B.Tech. (Civil Eng.)

Subject: Structural Dynamics.

Sub Code: PEC- CE410

Date: 15-10-2025

Time: 14:30 am to 15:15

Max. Marks: 20

Note:

- i. Assume suitable data whenever necessary and state it clearly.
- ii. Non programmable calculator is permitted.
- iii. Figures to right indicate full marks.
- iv. The symbols carry usual meaning.

Q.N.		Question	Marks	CO	BT
Q	1	Define and explain Time History Analysis in structural dynamics. Discuss, with reasons, why it is considered more accurate than Response Spectrum Analysis in seismic design.	06	CO3	BT2
Q	2	Calculate the displacement x at time $t = 0.1$ sec using the Newmark-beta method with parameters $m = 500$ kg, $k = 20000$ N/m, $\zeta = 0.05$, $F(t) = 1000 \sin(10t)$ N, $x(0) = 0$, $\dot{x}(0) = 0$, $\Delta t = 0.1$ sec, $t = 0.1$ sec $\beta = 0.25$, $\gamma = 0.5$	06	CO3	BT4
Q	3	Derive the expression for logarithmic decrement δ for a viscously damped free vibration system in terms of damping ratio ζ and natural frequency ω_n .	08	CO4	BT3

== ALL THE BEST ==

Course Outcomes:

CO 1	To recall fundamentals of Mathematics and Engineering Mechanics
CO 2	To Understand the effects of system/model parameters on dynamic response.
CO 3	To apply and formulate dynamic equilibrium equations for SDOF and MDOF system.
CO 4	To analyze SDOF and MDOF systems using classical and numerical methods.
CO 5	To evaluate and obtain the response of SDOF, MDOF systems and conduct modal analysis of MDOF Systems.

Bloom's Taxonomy (BT)

BT1-Remember, BT2- Understand, BT3- Apply, BT4- Analyse, BT5- Evaluate, BT6-Create



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Civil Engineering
ISE-II (Semester - I)
Academic Year: 2025-26

Class: Final Year-B.Tech.

Subject: Elective: Construction Engineering & Management

Subject Code: PEC-CE408

Date: 15/10/2025

Time: 14:30 pm to 15:15 pm

Max. Marks: 20

Note:

- i. Non-programmable calculator is permitted.
- ii. Figures to right indicate full marks.
- iii. Attempt any ONE (a OR b) from Q No. 2.

iii. Attempt any ONE (a OR b) from Q No 2.

Q.No.	Question		Marks	CO	BT																														
Q1	a	List and Explain any two earth-moving equipments with neat sketches.	06	CO1	BT2																														
Q2	a	For the construction of guest house, certain activities are to be performed which are as given below: <table><tr><th>Activity</th><th>A</th><th>B</th><th>C</th><th>D</th><th>E</th><th>F</th><th>G</th></tr><tr><td>Duration (in weeks)</td><td>2</td><td>3</td><td>5</td><td>4</td><td>2</td><td>3</td><td>5</td></tr></table> Activity B and C can be performed simultaneously and can start only when activity A is completed. Activity D can start only after activity B ends. Activity E cannot begin until activities B & C are completed. Activity F can start only after activities D & E are complete, activity G is the last activity and this can commence only after the completion of activity E. Prepare a bar chart for the project. What is the total time taken for the completion of the project? Allocate resources such that contractor should not use more than 6 labours a day. Draw final resource profile.	Activity	A	B	C	D	E	F	G	Duration (in weeks)	2	3	5	4	2	3	5	08	CO3	BT3, 4														
Activity	A	B	C	D	E	F	G																												
Duration (in weeks)	2	3	5	4	2	3	5																												
	b	The optimistic, likely and pessimistic time estimates for the PERT network of a project are shown in table below. Calculate standard deviation and variance of the project: <table><tr><th>Activity</th><th>IPA</th><th>Optimistic Time (days)</th><th>Most likely time (days)</th><th>Pessimistic time (days)</th></tr><tr><td>A</td><td>—</td><td>3</td><td>4</td><td>8</td></tr><tr><td>B</td><td>A</td><td>4</td><td>6</td><td>8</td></tr><tr><td>C</td><td>B</td><td>7</td><td>7</td><td>8</td></tr><tr><td>D</td><td>B, C</td><td>4</td><td>5.5</td><td>7</td></tr><tr><td>E</td><td>D</td><td>6</td><td>7</td><td>9</td></tr></table>	Activity	IPA	Optimistic Time (days)	Most likely time (days)	Pessimistic time (days)	A	—	3	4	8	B	A	4	6	8	C	B	7	7	8	D	B, C	4	5.5	7	E	D	6	7	9	08	CO3	BT3, 4
Activity	IPA	Optimistic Time (days)	Most likely time (days)	Pessimistic time (days)																															
A	—	3	4	8																															
B	A	4	6	8																															
C	B	7	7	8																															
D	B, C	4	5.5	7																															
E	D	6	7	9																															
Q3	a	Explain stepwise modular construction method with a case study on “Mini Sky City, China (57-storey building)”.	06	CO2	BT5																														

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyse, BT5 - Evaluate, BT6 -

Create

Course Outcomes:

CO	On successful completion of this course, students should be able to:
CO1	Define the fundamental principles of management theories
CO2	Understand the feasibility analysis in construction management and network analysis tools for cost and time estimation
CO3	Apply theoretical and practical aspects of construction management techniques to achieve project goals
CO4	Analyze contemporary construction management tools and methodologies in Indian context.



Note: Mobile phones and programmable calculators are strictly prohibited.

Q.N.	Questions	Marks	CO	BT
Q.1	Find the value of ultimate bearing capacity for a square footing of 1.5 m wide placed at a depth of 2 m below the ground on a cohesive soil. The unconfined compressive strength of soil is 20 kN/m^2 and unit weight is 17.5 kN/m^3 . Use general shear failure and Meyerhof's analysis.	05	4	4
Q.2	Explain with neat sketch all types of bearing capacity failures.	05	2	2
Q.3	Explain the impact of moisture changes in soil for all types of construction.	05	2	2
Q.4	Illustrate the preventive measures to build foundation on expansive soil.	05	3	3

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Q.3	a	What is an unconformity? Explain any two types of unconformities with neat sketches.	04	CO3	BT2
	b	Describe any three types of folds and explain their engineering importance.	05	CO3	BT2

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 – Remember BT2 – Understand BT3 – Apply BT4 – Analyse BT5 – Evaluate BT6 – Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Explain the weathering process and mass movement.
CO 2	Distinguish geological formations
CO 3	Identify geological structures and processes for rock mass quality
CO4	Identify subsurface information and groundwater potential sites through geophysical investigations
CO5	Apply geological principles for mitigation of natural hazards and select sites for dams and tunnels